Chapter 3: Functional group protection

General considerations

Main features of a protecting group

Compatibility and orthogonality between protecting groups

Classification of protecting groups according to the cleavage conditions

Functional group protection

Protection of hydroxyl, carbonyl, carboxyl and amino groups

References:

- ✓ F.A. Carey, R.J. Sundberg, *Advanced Organic Chemistry. Part B: Reactions and Synthesis*, 4ª ed., Kluwer, 2001. Capítol 13.1
- ✓ T.W. Greene, P.G.M. Wuts, *Protective Groups in Organic Synthesis*, 3ª ed.. Wiley, 1999
- ✓ P.J. Kocieński, *Protecting Groups*, Thieme, 1994.
- ✓ J. Robertson, *Protecting Group Chemistry*, Oxford Chemistry Primers, 2000

Universitat de Girona

3.1. General considerations

- Issues that define the suitability of a protecting group:

  a) It should be cheap or readily available

  b) It should be easily and efficiently introduced

  c) It should be easy to characterize and avoid such complications as the creation of new stereogenic centers

  d) It should be stable to the widest possible range of reaction conditions and to the work-up steps

  e) It should be stable to the different separation and purification techniques, such as chromatography

  f) It should be removed selectively and efficiently under highly specific conditions

  g) The by-products of the deprotection should be easily separated from the substrate

- Two protecting groups are compatible if they show a gradual lability towards a given reagent.

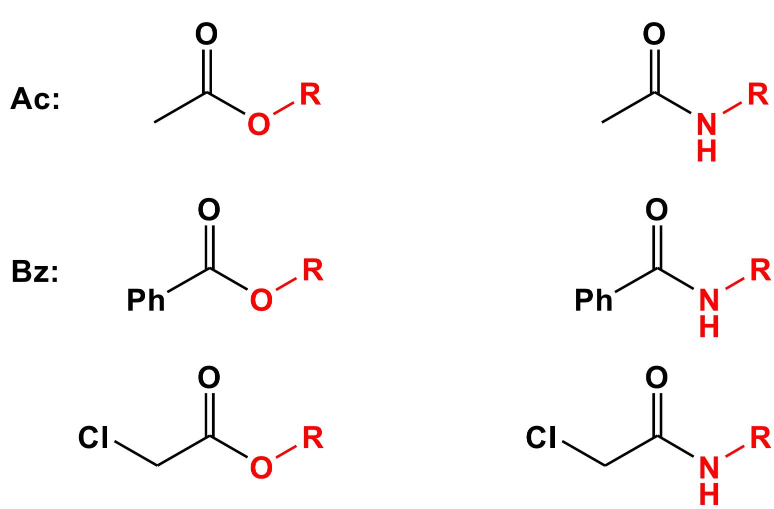
- Two protecting groups are orthogonal if one can be removed in the presence of the other and in any order. Their removal mechanisms are different.

3.2. Classification of protecting groups according to the cleavage conditions

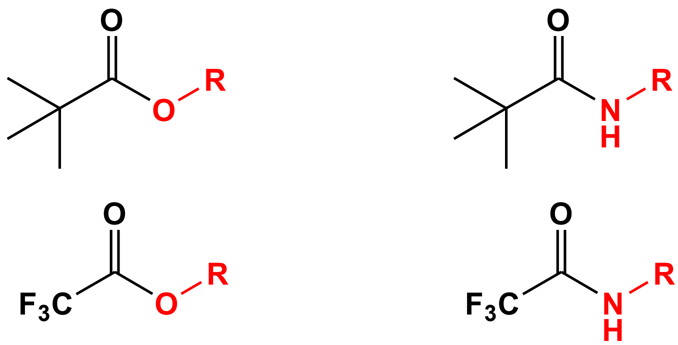
3.2.1. Protecting groups cleaved by basic solvolysis
3.2.2. Protecting groups cleaved by acid
3.2.3. Protecting groups cleaved by a heavy metal
3.2.4. Protecting groups cleaved by fluoride ions
3.2.5. Protecting groups cleaved by reductive elimination
3.2.6. Protecting groups cleaved by β-elimination reactions
3.2.7. Protecting groups cleaved by hydrogenolysis
3.2.8. Protecting groups cleaved by oxidation
3.2.9. Protecting groups cleaved by dissolving metal reduction
3.2.10. Protecting groups cleaved by transition metal catalysis
3.2.11. Protecting groups cleaved by light

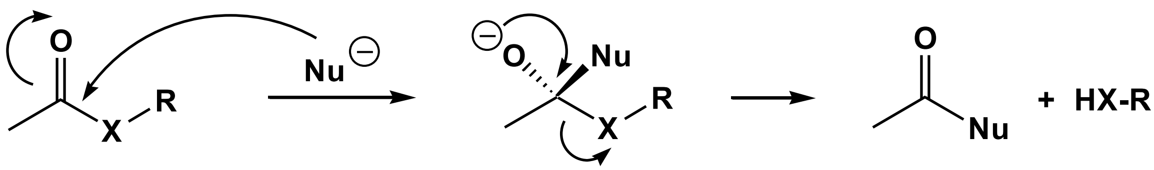
3.2.1. Protecting groups cleaved by basic solvolysis

Acyl derivatives of hydroxyl and amino groups:



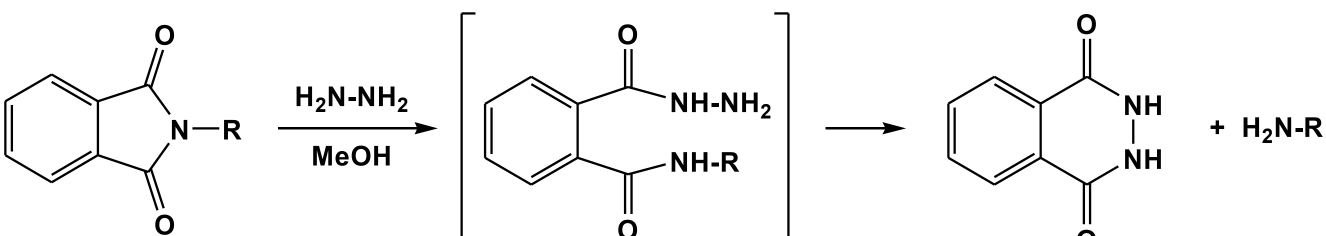
Pv:





X= O, NH

Phtaloyl group, a special case:



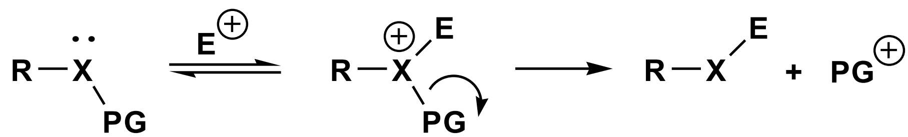
Pht-N-R

3.2.2. Protecting groups cleaved by acid

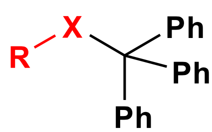
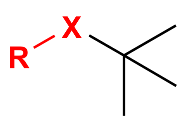
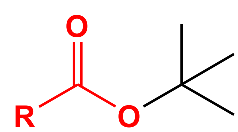
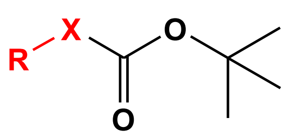
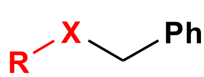
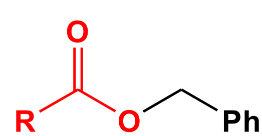
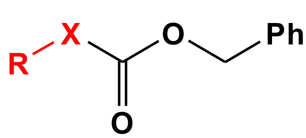
They can be divided into two subsets:

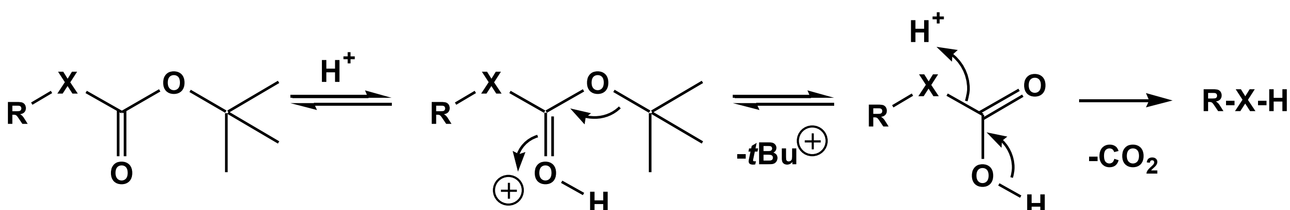
- Tertiary alkyl or benzylic amines, ethers, esters, carbonates and carbamates
- *O,O*-acetals

- Tertiary alkyl or benzylic amines, ethers, esters, carbonates and carbamates



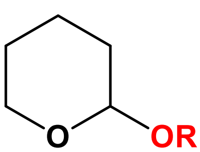
X= O, NH, S
E^+= H^+ or Lewis acid

Trt:		**X= O, NH, S**		**Labile to AcOH**
***t*Bu:**		**X= O, NH**		**Labile to TFA**
Boc:		**X= O, NH**		**Labile to TFA**
Bn:		**X= O, NH, S**		**Labile to HBr or HF**
Z:		**X= O, NH**		**Labile to HBr or HF**

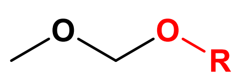


X= O, NH

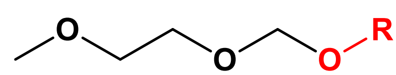
- *O,O*-acetals



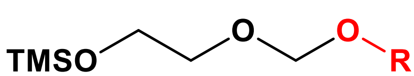
THP



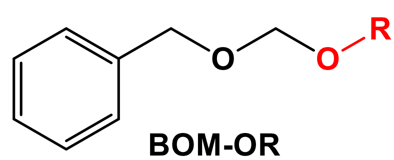
MOM-OR



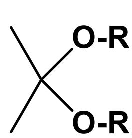
MEM-OR

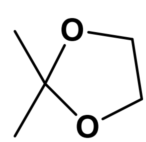


SEM-OR



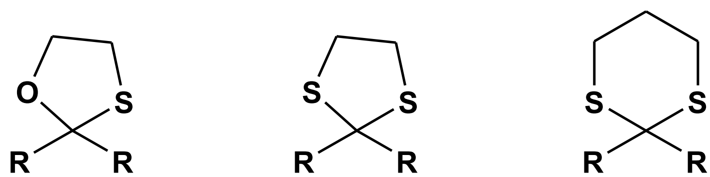
BOM-OR



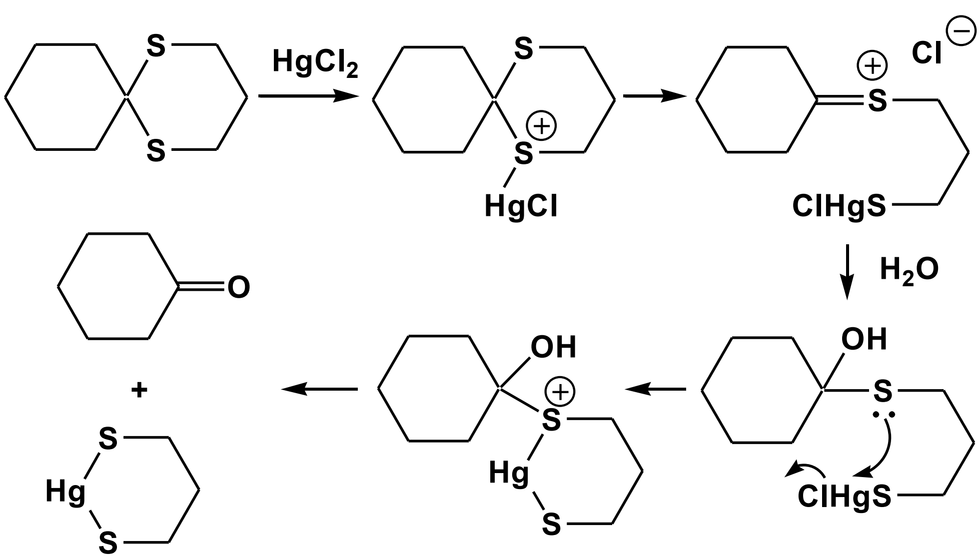


3.2.3. Protecting groups cleaved by a heavy metal

O,S-acetals *S,S*-acetals

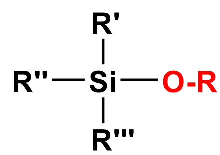


oxathiolane **dithiolane** **dithiane**



3.2.4. Protecting groups cleaved by fluoride ions

➢Cleavage of Si-O bonds: silyl ethers



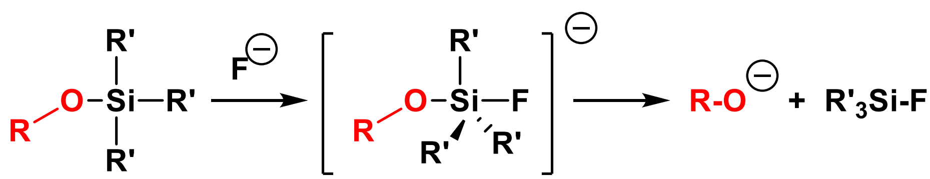
Acid catalyzed solvolysis:

Me_3Si- < Et_3Si- < *t*$BuMe_2Si$- < *i*Pr_3Si- < *t*$BuPh_2Si$-

TMS TES TBDMS or TBS TIPS TBDPS

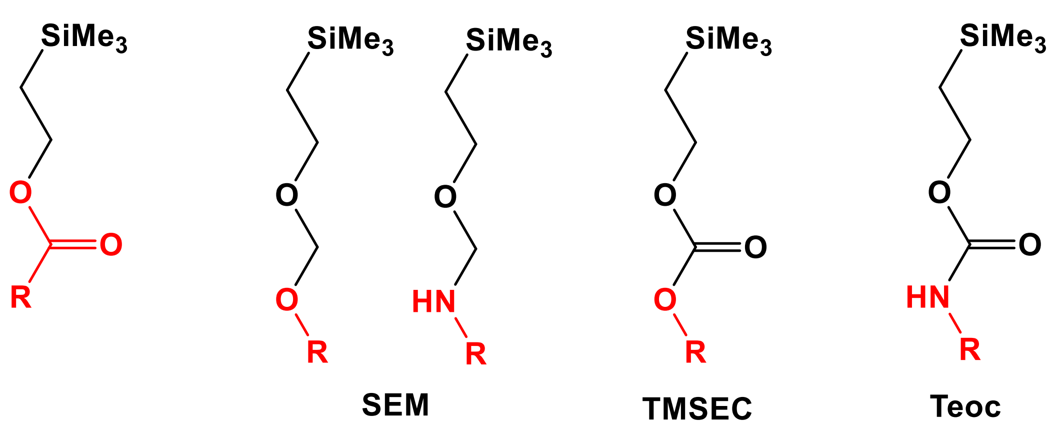
Base catalyzed solvolysis:

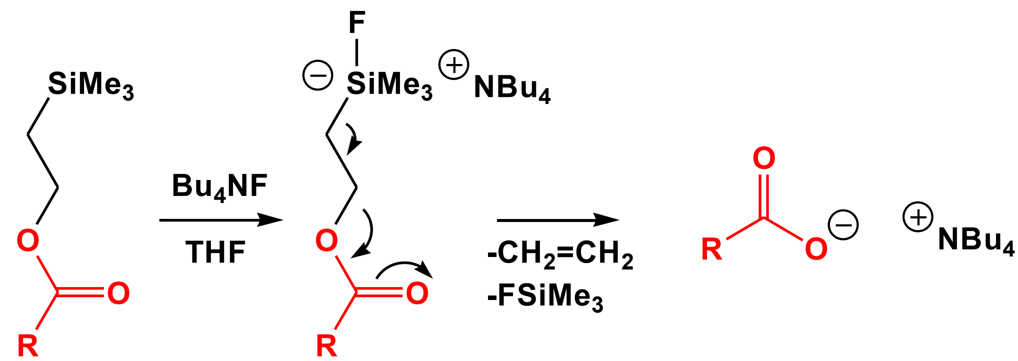
Me_3Si- < Et_3Si- < *t*$BuMe_2Si$- ≈ *t*$BuPh_2Si$- < *i*Pr_3Si-



Reagents: *n*$Bu_4N^+F^-$ (TBAF) in THF or HF in MeCN

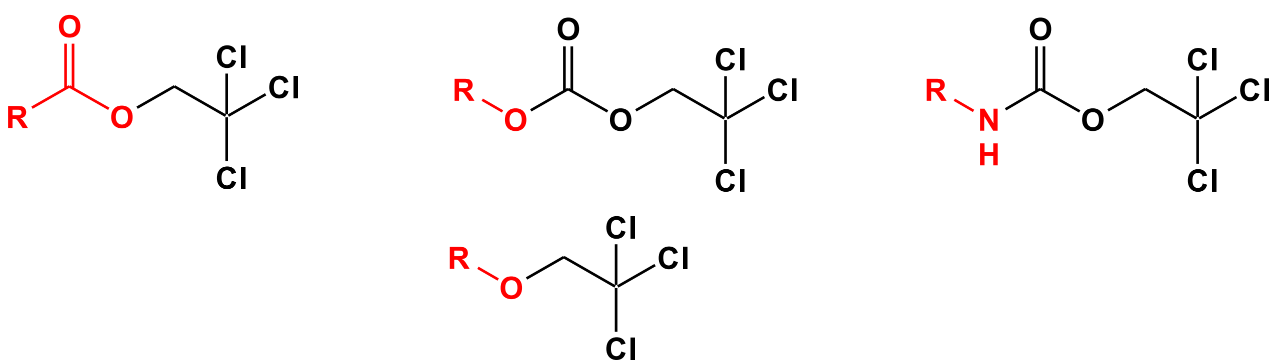
➢Cleavage of Si-C bonds

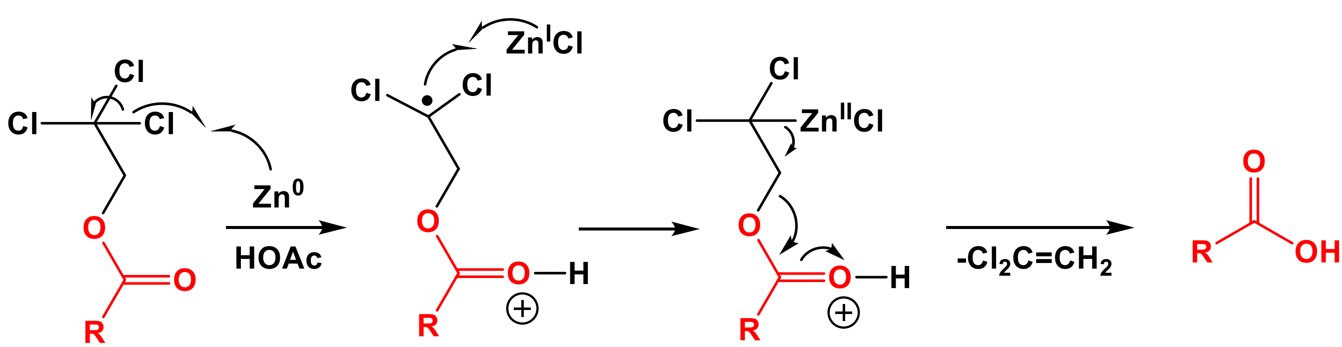




3.2.5. Protecting groups cleaved by reductive elimination

2,2,2-Trichloroethyl derivatives:

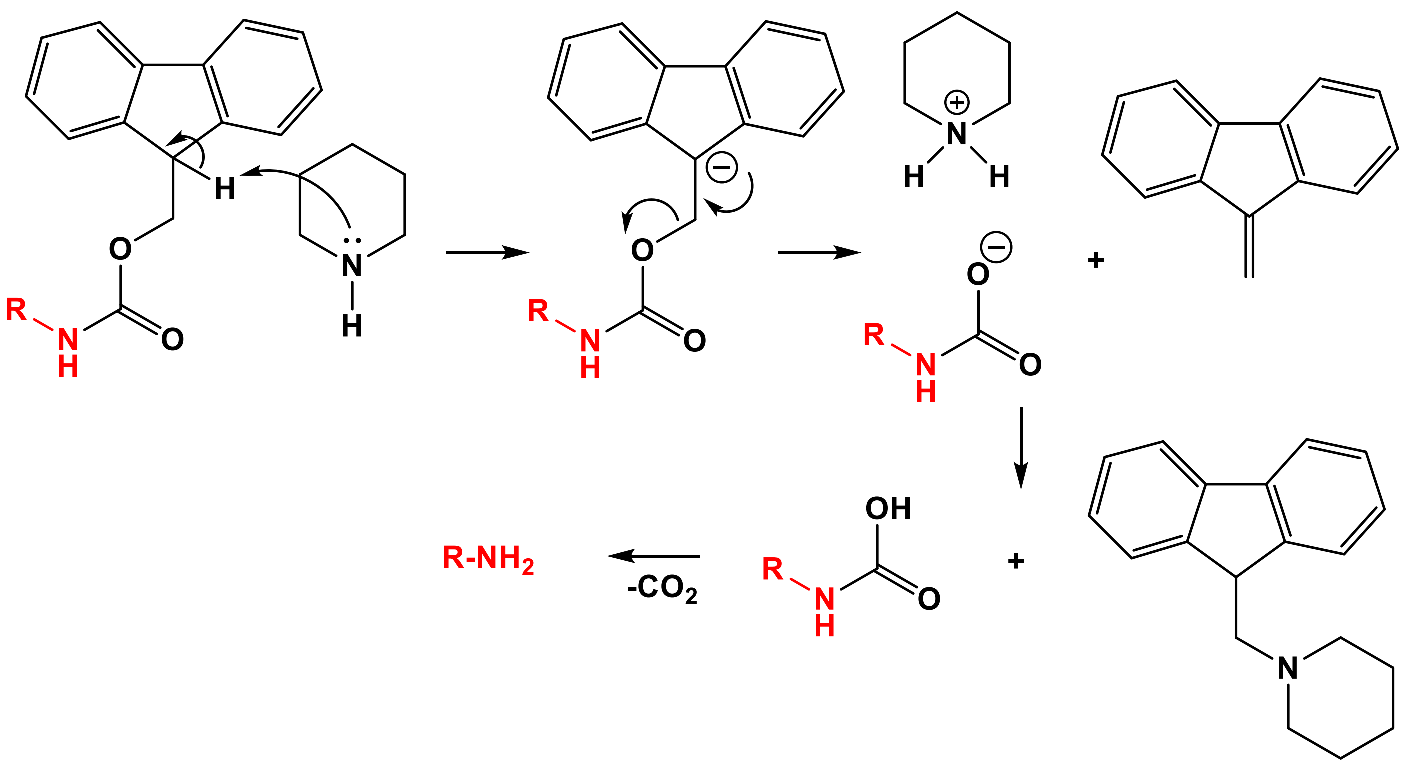




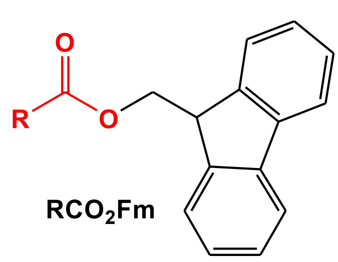
3.2.6. Protecting groups cleaved by β-elimination reactions

An important example of this subset is the 9-fluorenylmethoxycarbonyl (Fmoc) group.

It is commonly removed with secondary amines, such as piperidine, through a E1cB mechanism. Piperidine removes the acidic proton at the C-9 position of the fluorene ring. This proton is acidic because the resulting anion is stabilized by the two aromatic rings. Dibenzofulvene is released, which reacts with piperidine forming a stable adduct that can be separated from the friee amine.

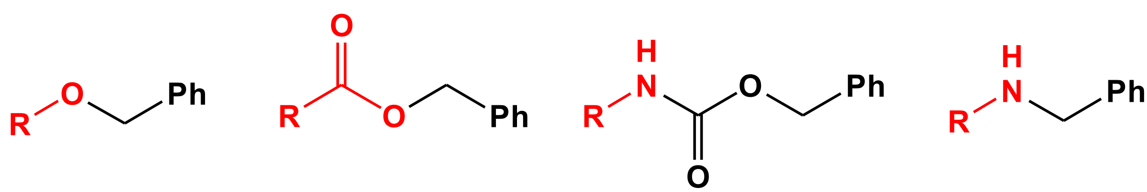


The 9-fluorenylmethyl (Fm) group is applied to the protection of carboxyl groups:

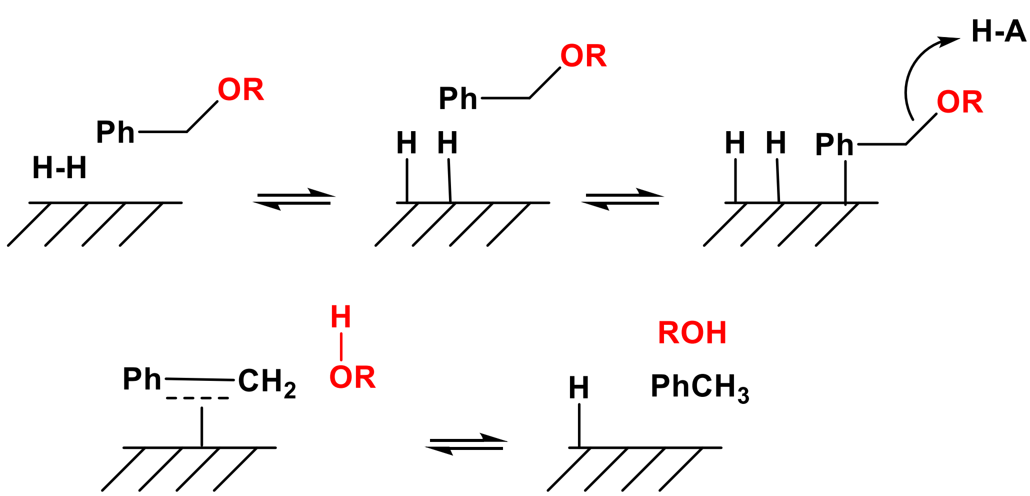


3.2.7. Protecting groups cleaved by hydrogenolysis

An excellent method for cleaving benzylic ethers, esters, carbamates, and amines

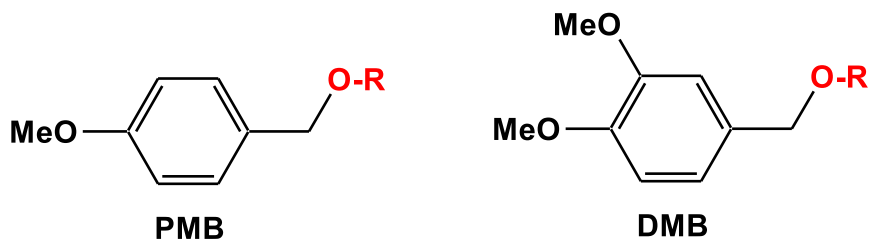


It uses H_2 in the presence of a transition metal such as Pd



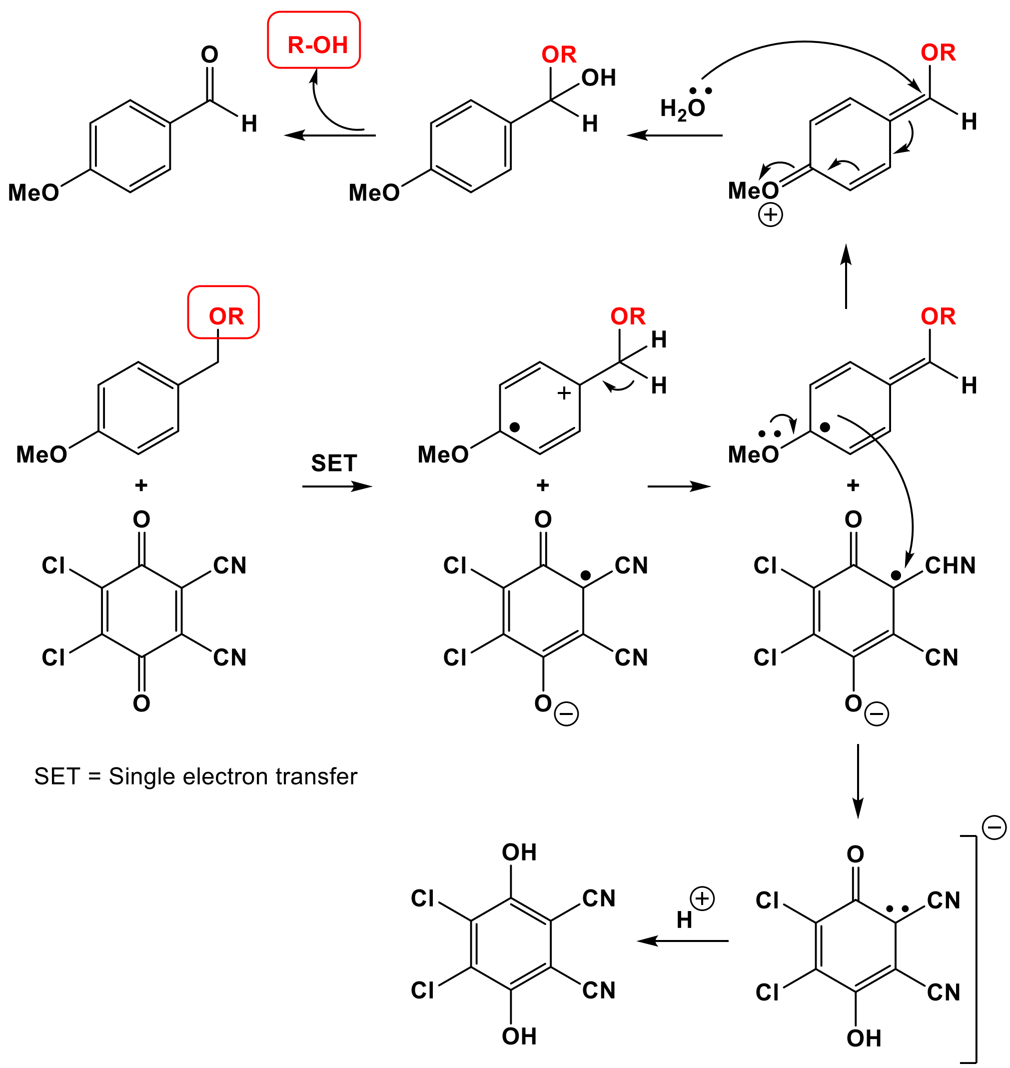
Alternatively a process known as *catalytic transfer hydrogenation* can be employed. It uses 1,4-cyclohexadiene, cyclohexene, formic acid or ammonium formate as H_2 source.

3.2.8. Protecting groups cleaved by oxidation



Oxidant reagents:

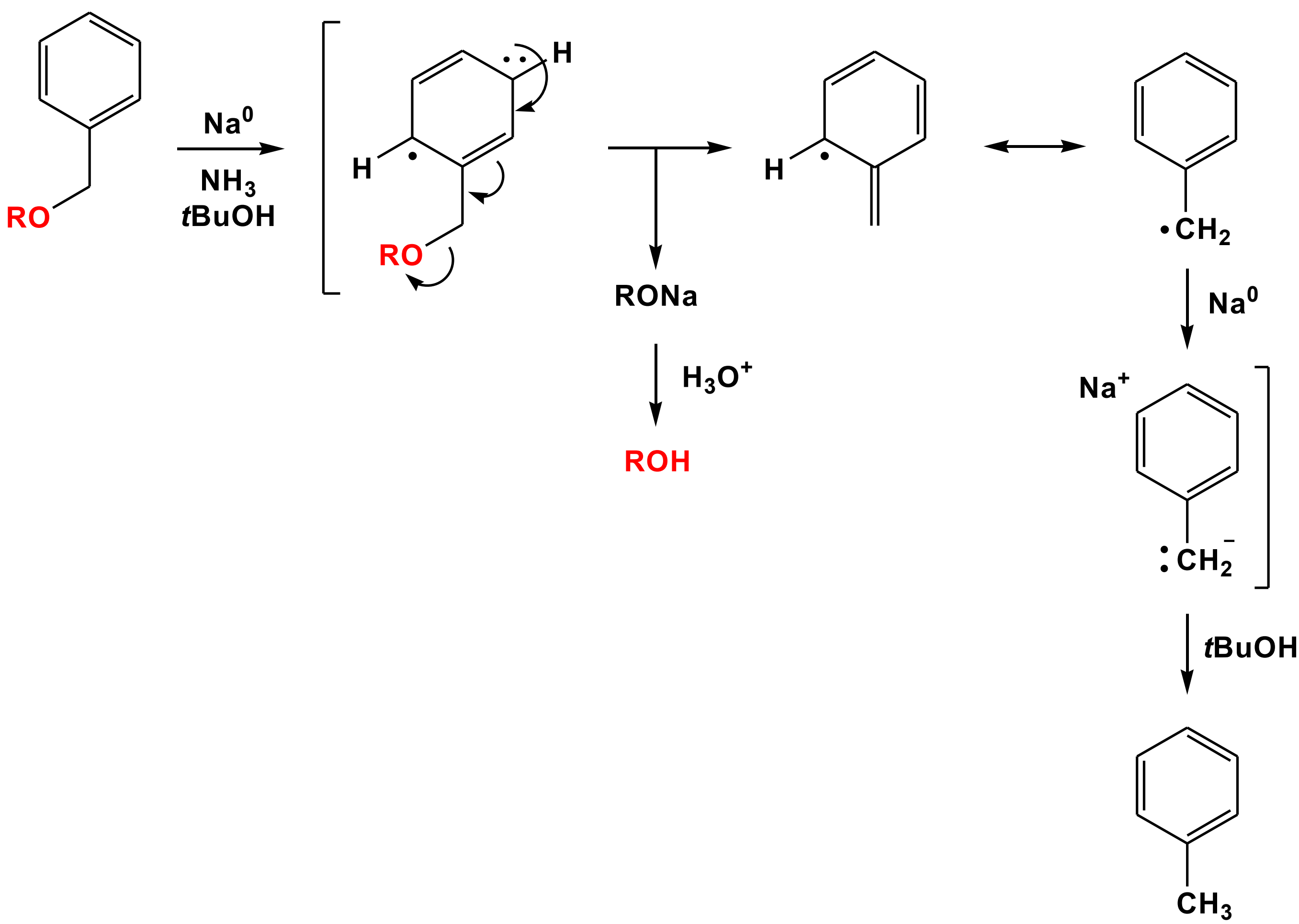
2,3-Dichloro-5,6-dicyano-1,4-benzoquinone (DDQ)
$(NH_4)_2Ce(NO_3)_6$ (CAN)



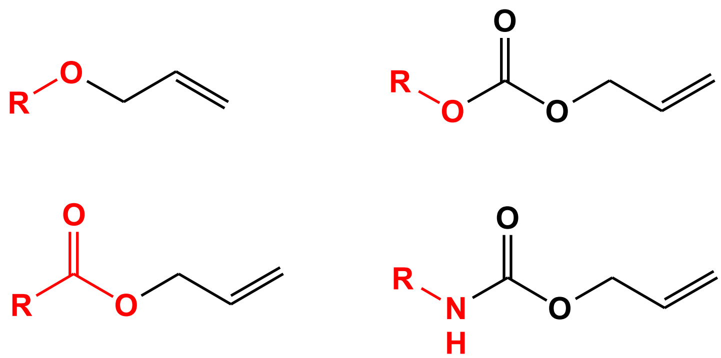
SET = Single electron transfer

3.2.9. Protecting groups cleaved by dissolving metal reduction

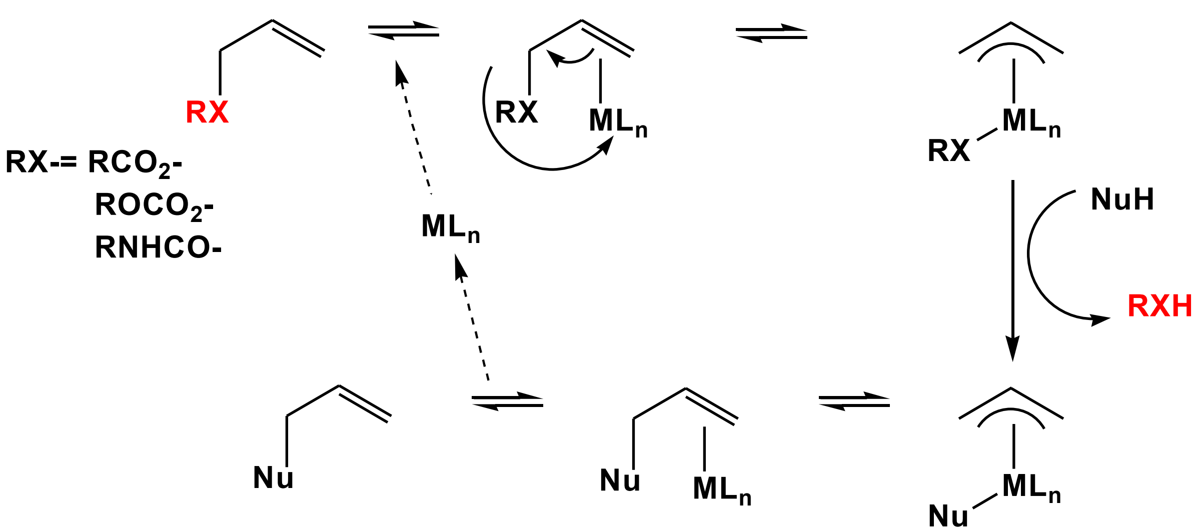
Benzylic ethers and esters. Conditions: Na or Li in liquid NH_3



3.2.10. Protecting groups cleaved by transition metal catalysis

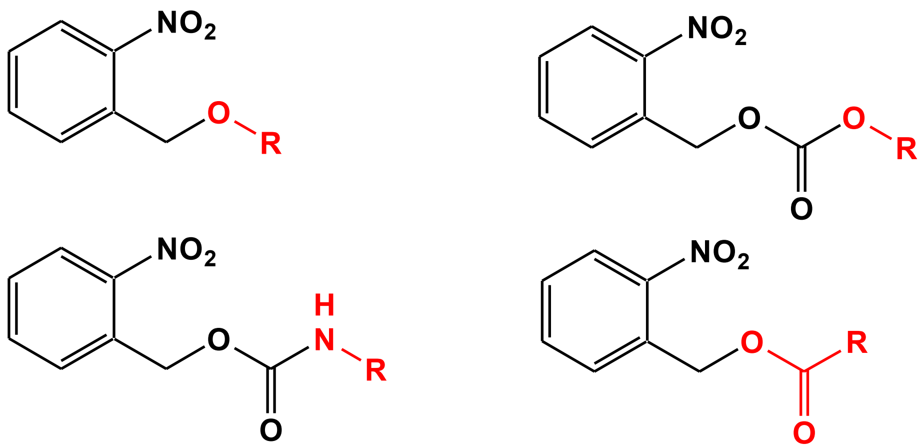


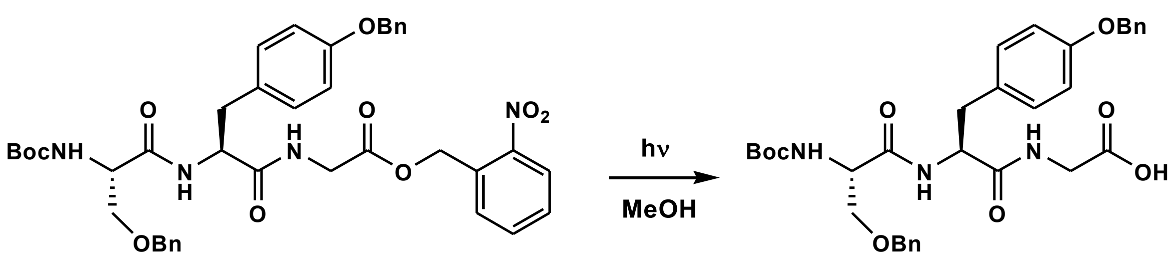
Conditions: $Pd(PPh_3)_4$, $PdCl_2$



NuH: Bu_3SnH, morpholine, AcOH

3.2.11. Protecting groups cleaved by light



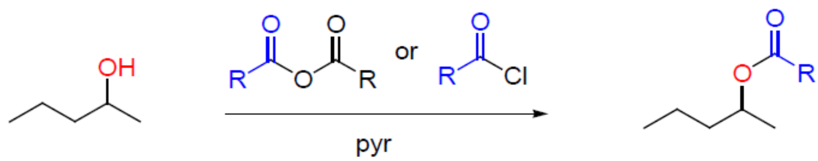


3.3. Functional group protection

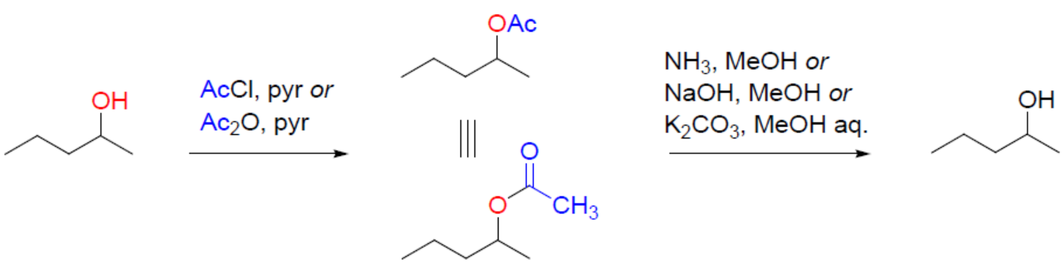
3.3.1. Hydroxyl protecting groups

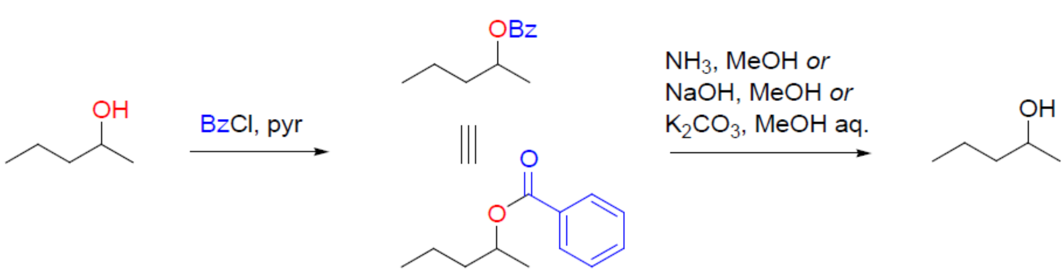
Esters

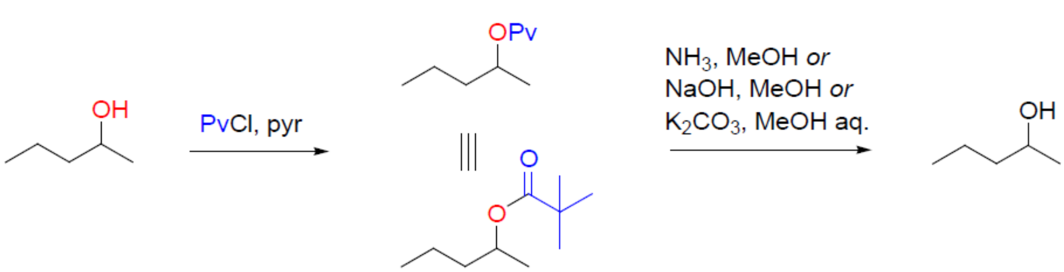
Formation:



Examples and cleavage conditions:



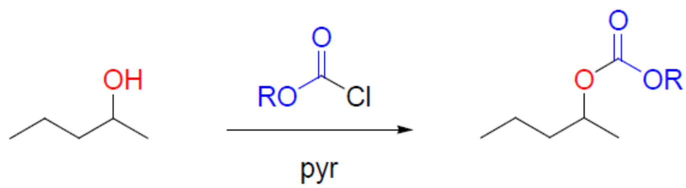




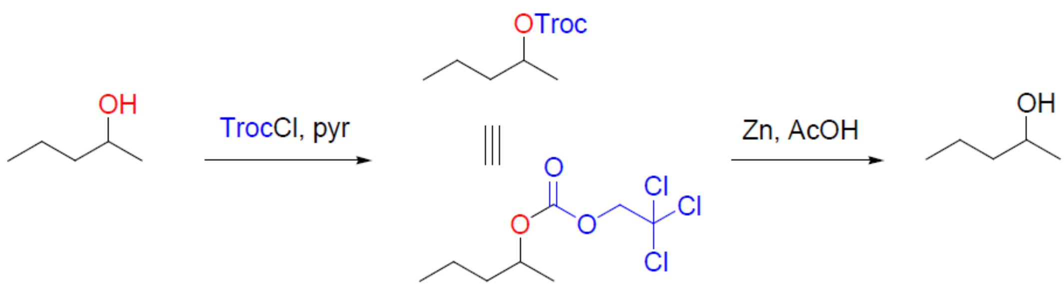
Reactivity order: Pv- < Bz- < Ac-

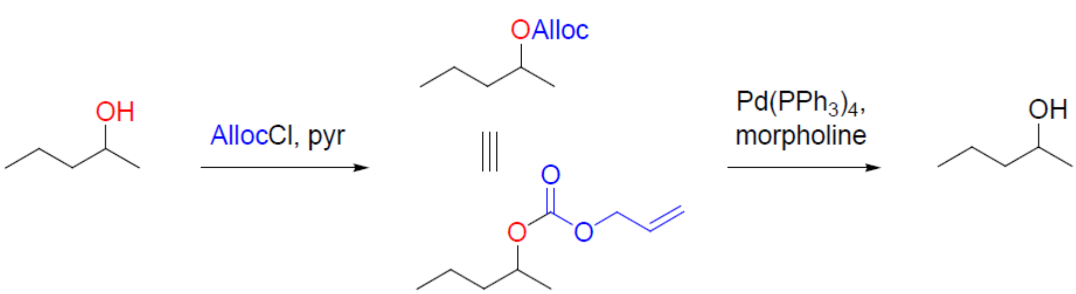
Carbonates

Formation:



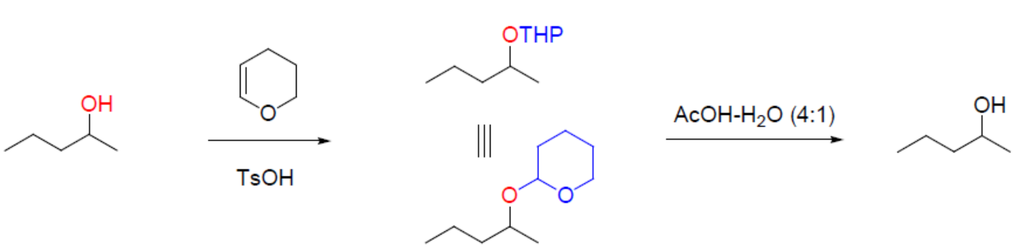
Examples and cleavage conditions:





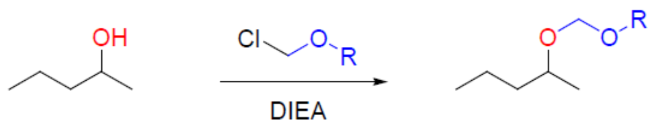
Ethers

✓ Tetrahydropyranyl

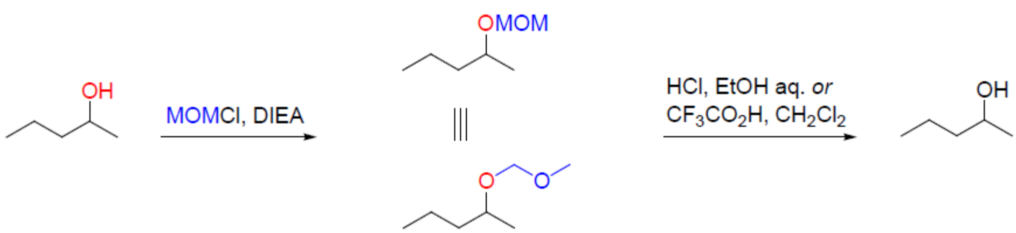


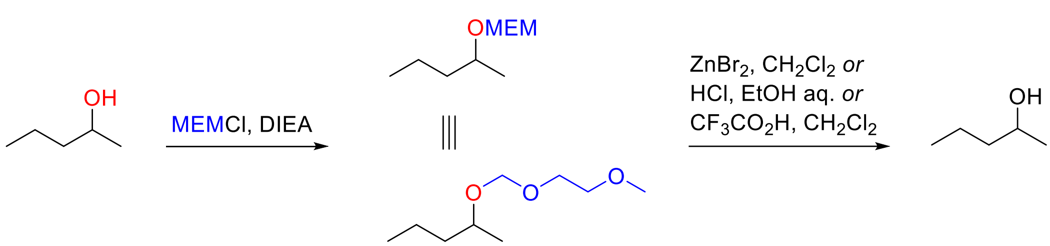
✓ Alkoxymethyl ethers

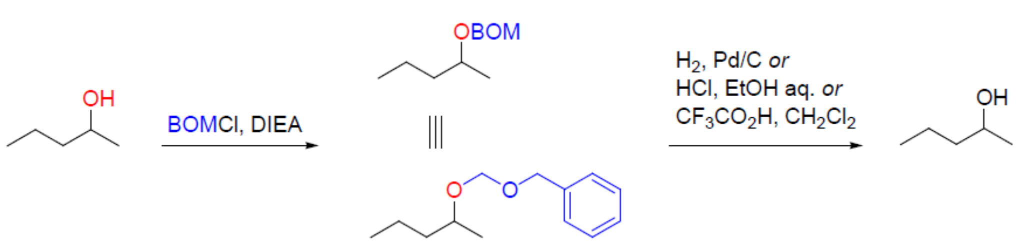
Formation:

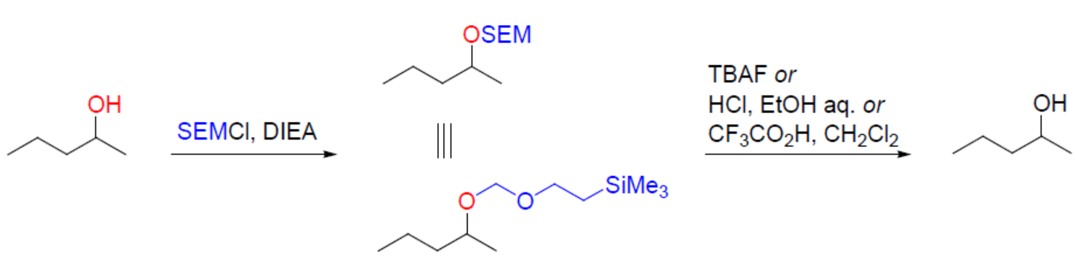


Examples and cleavage conditions:



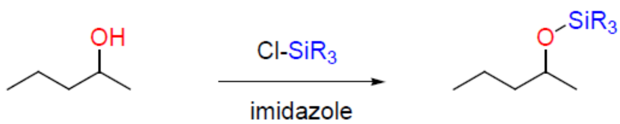




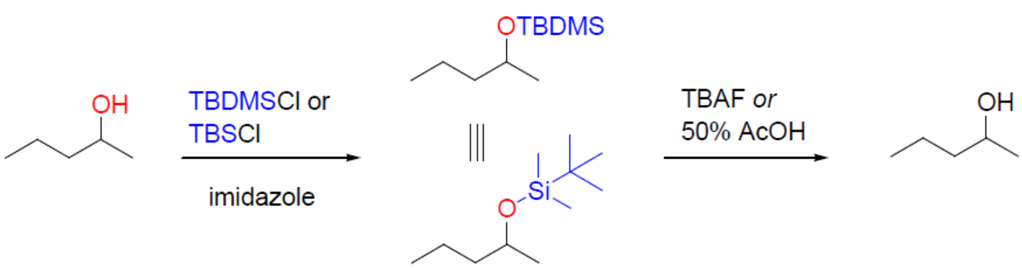


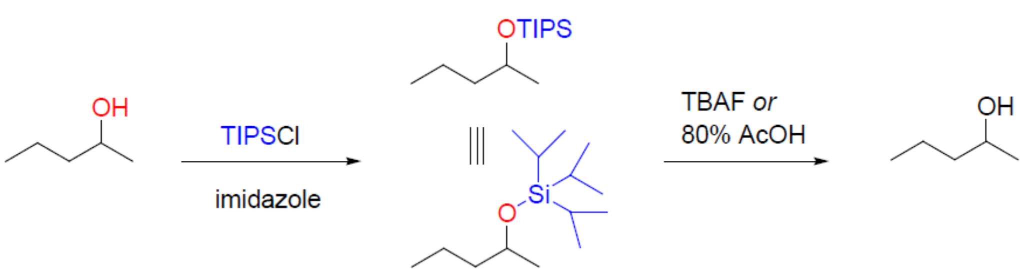
✓ Silyl ethers

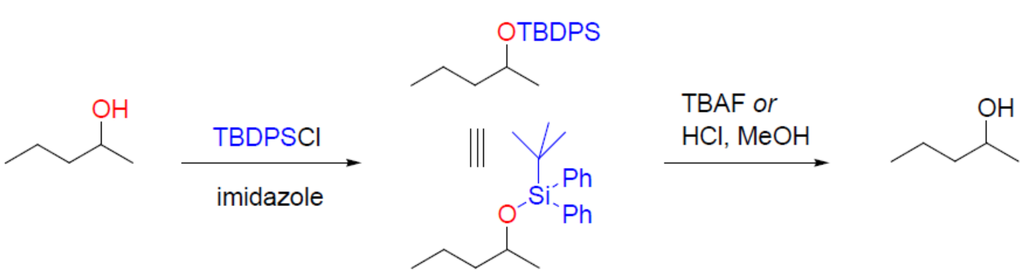
Formation:



Examples and cleavage conditions:



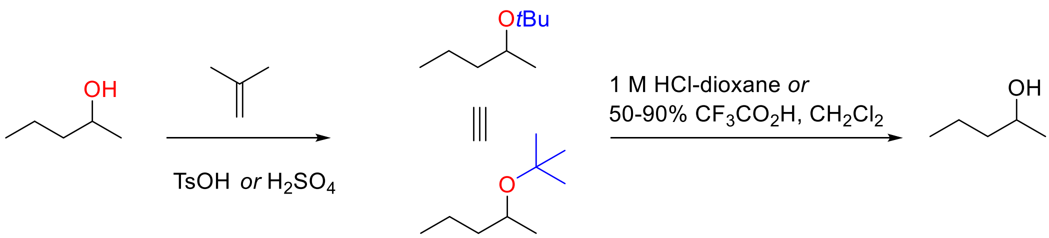


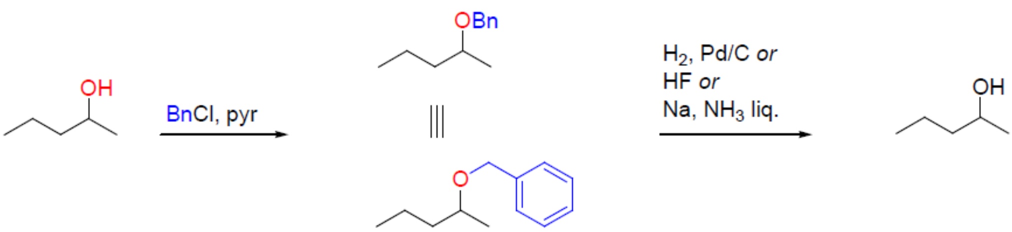


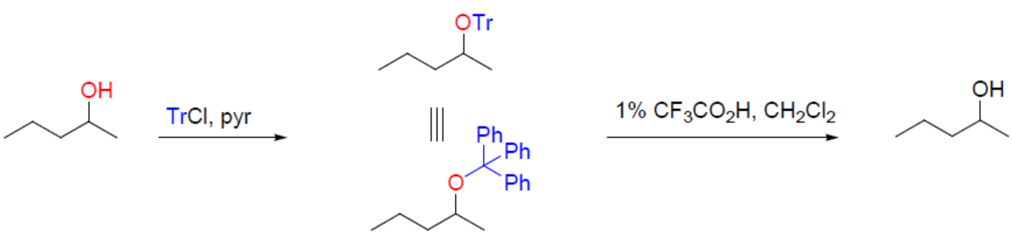
Chapter 3. Functional group protection

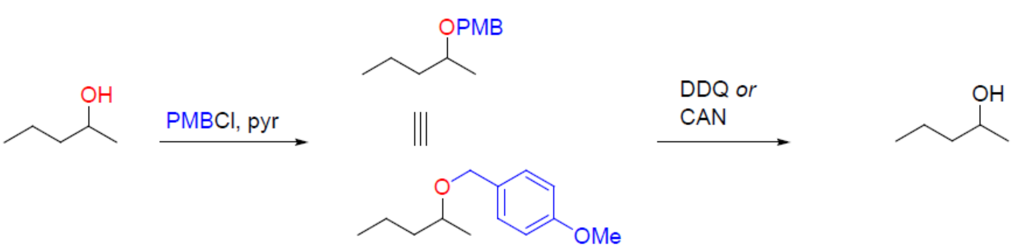
✓ Alkyl ethers

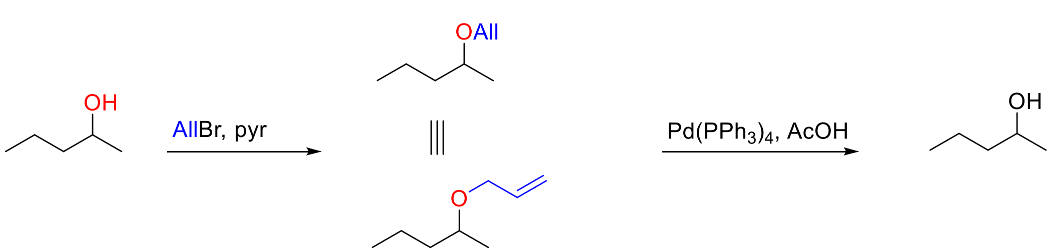
Examples, formation and cleavage conditions:







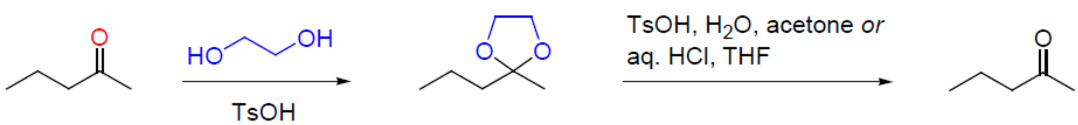


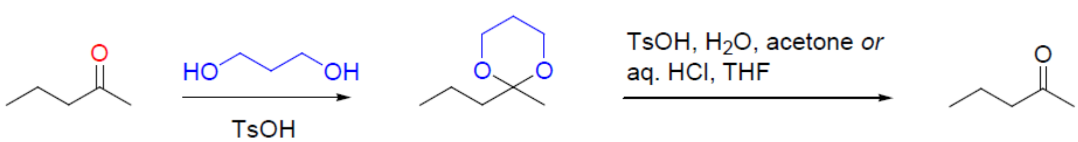


3.3.2. Carbonyl protecting groups

O,*O*-Acetals

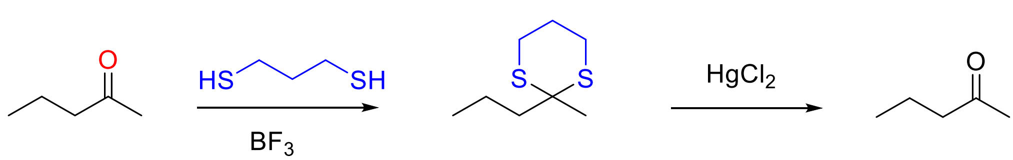
Formation and cleavage conditions:





S,*S*-Acetals

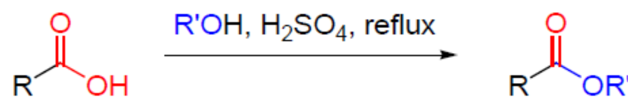
Formation and cleavage conditions:

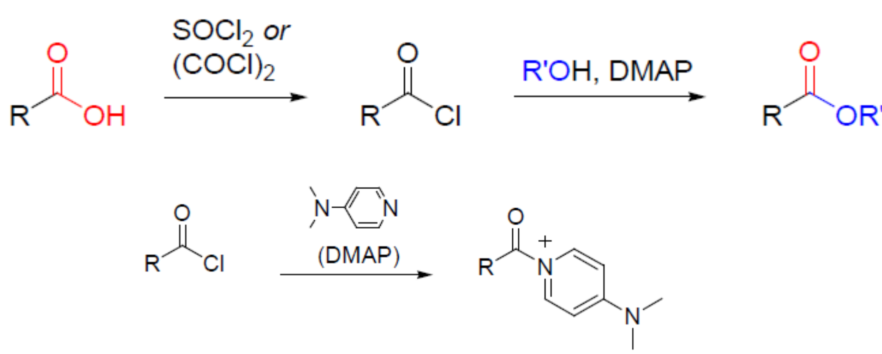


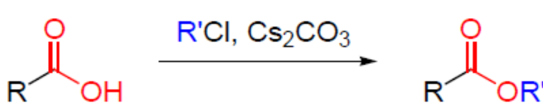
3.3.3. Carboxyl protecting groups

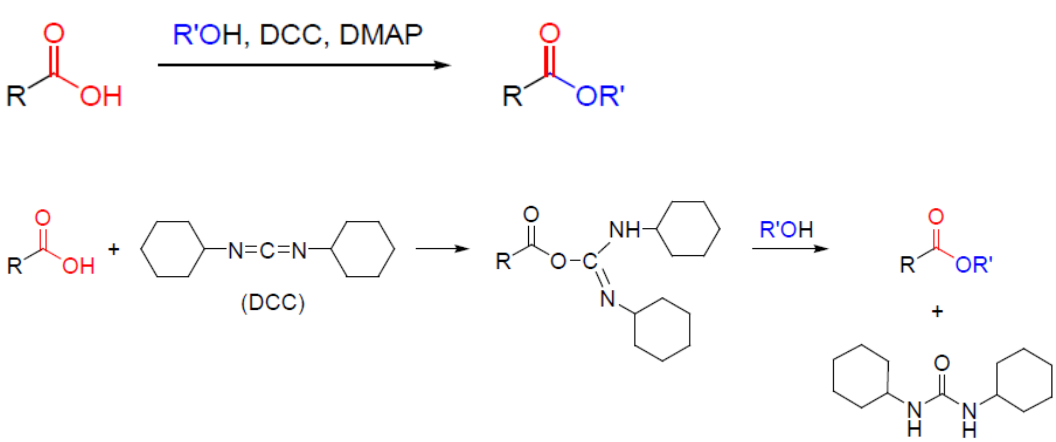
Esters

General esterification methods:

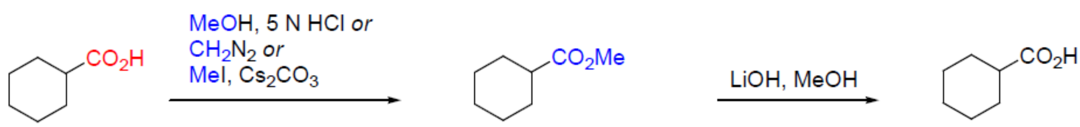


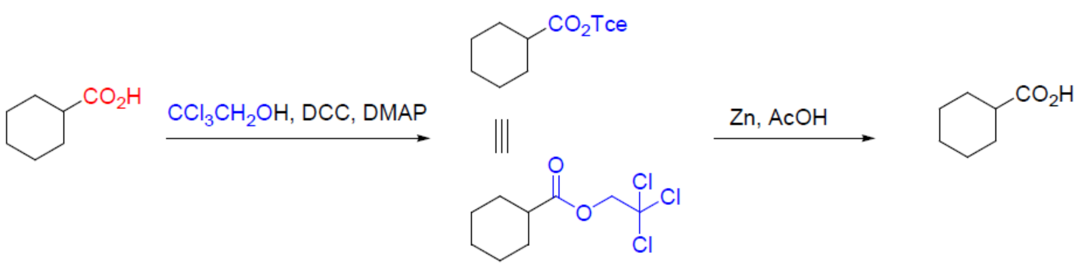


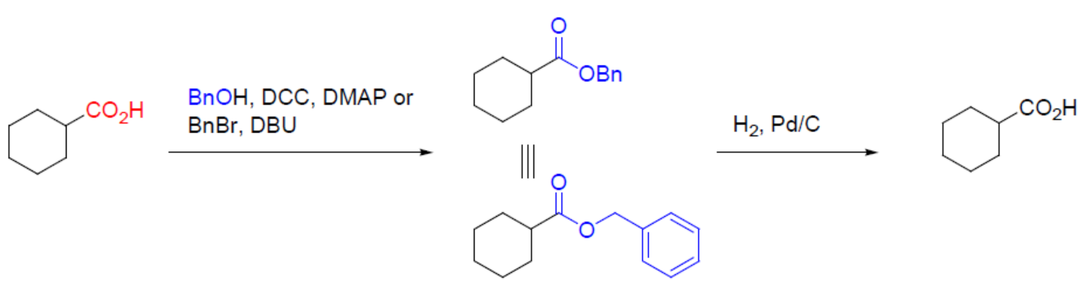


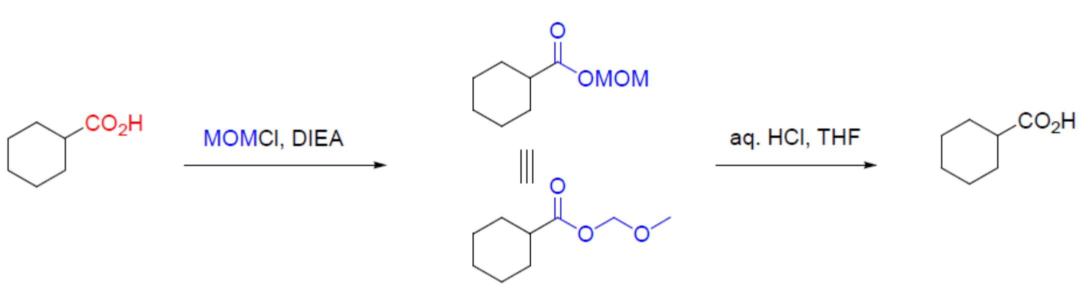


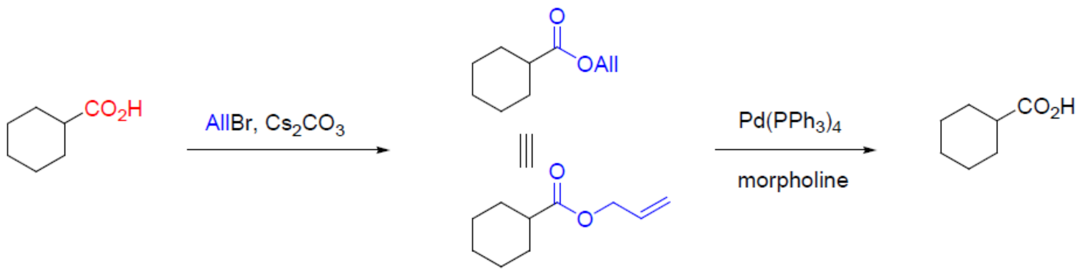
Examples, formation and cleavage conditions:

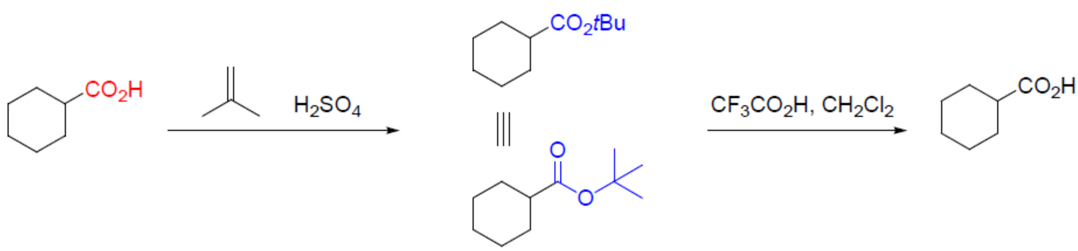


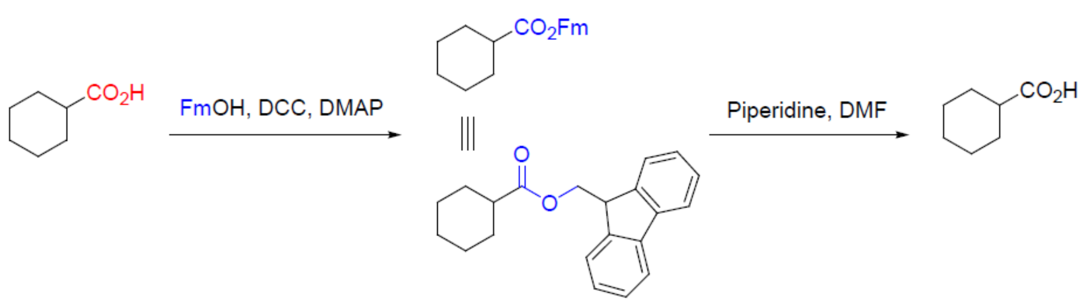


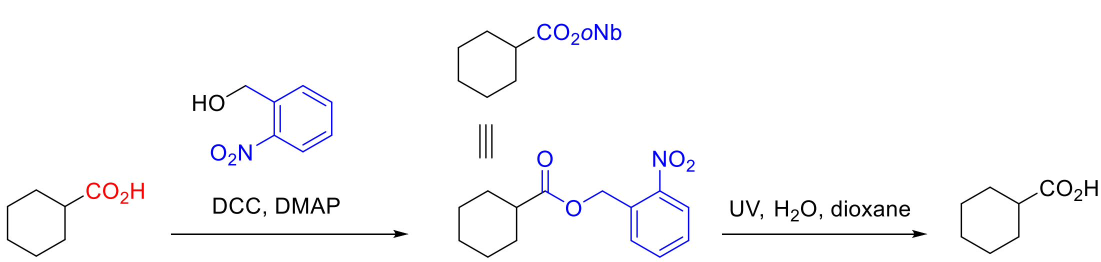





CO2H
AllBr, Cs2CO3
OAll
Pd(PPh3)4
morpholine
CO2H


CO2H
H2SO4
CO2tBu
CF3CO2H, CH2Cl2
CO2H


CO2H
FmOH, DCC, DMAP
CO2Fm
Piperidine, DMF
CO2H

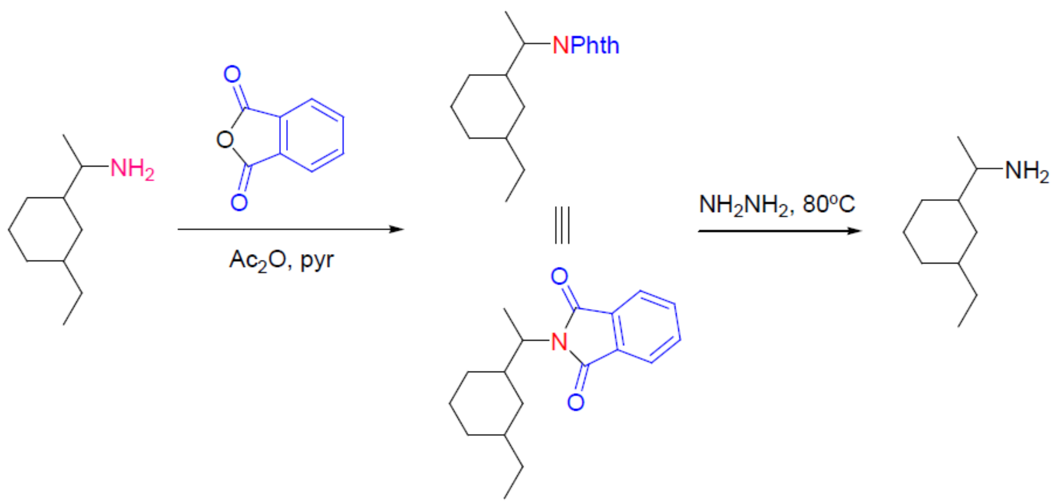

CO2H
HO
O2N
DCC, DMAP
CO2oNb
NO2
UV, H2O, dioxane
CO2H

3.3.4. Amino protecting groups

Imides

Protecting groups for primary amines

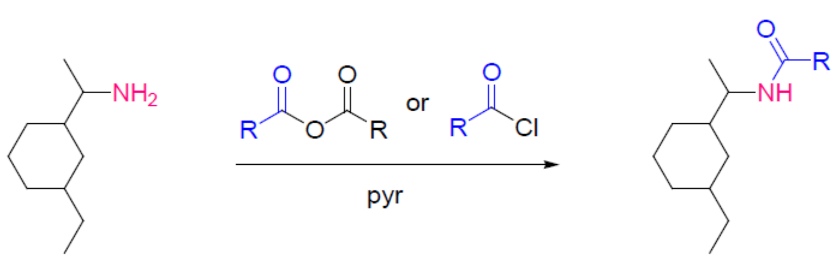
Formation and cleavage conditions:



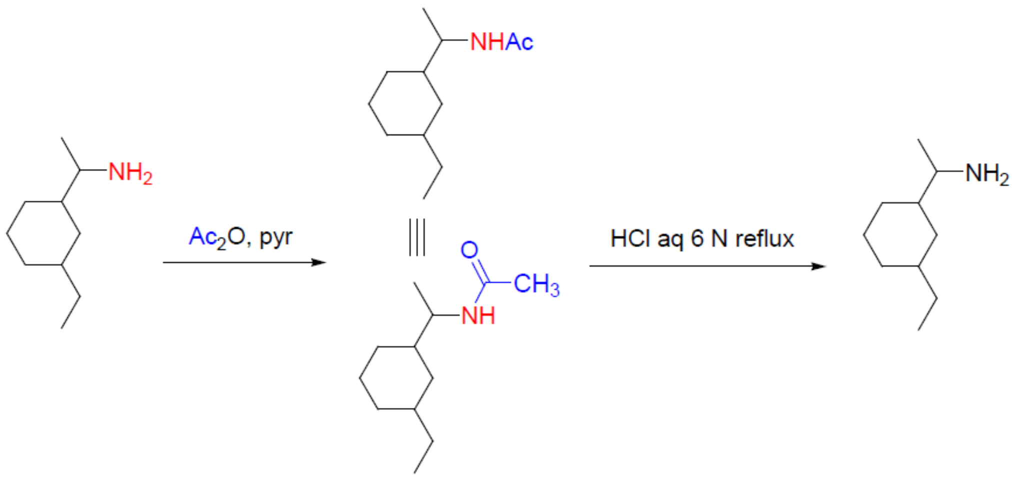
Amides

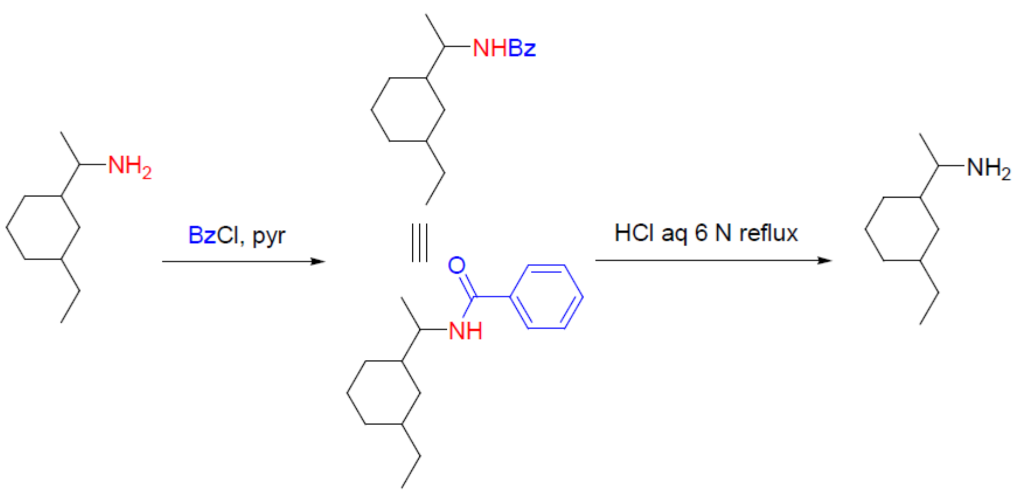
Protecting groups for primary and secondary amines

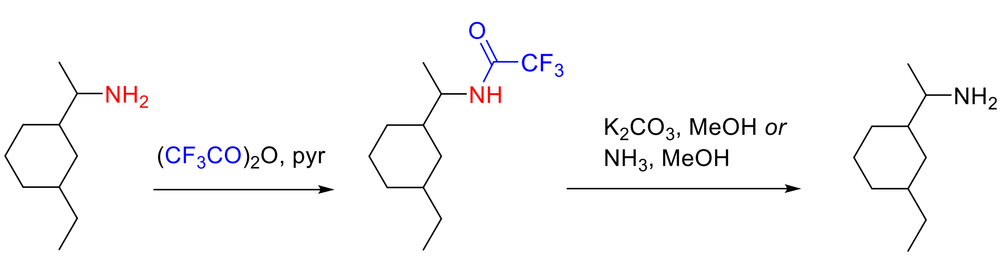
Formation:



Examples and cleavage conditions:



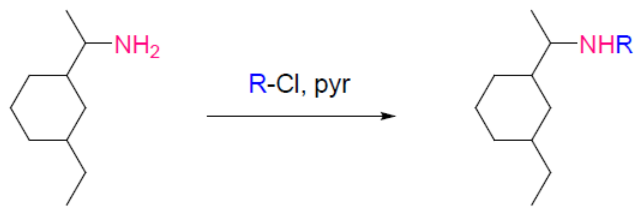




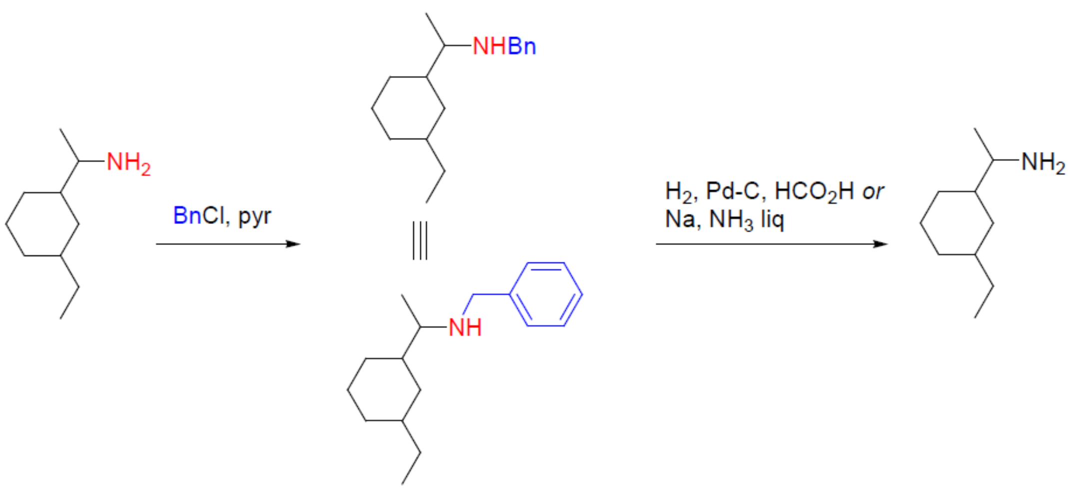
N-Alkyl derivatives

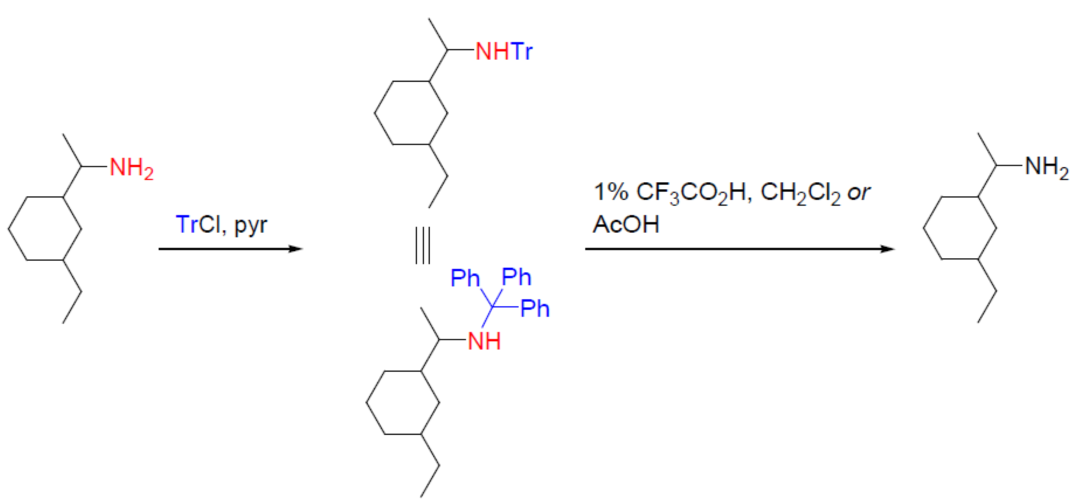
Protecting groups for primary and secondary amines

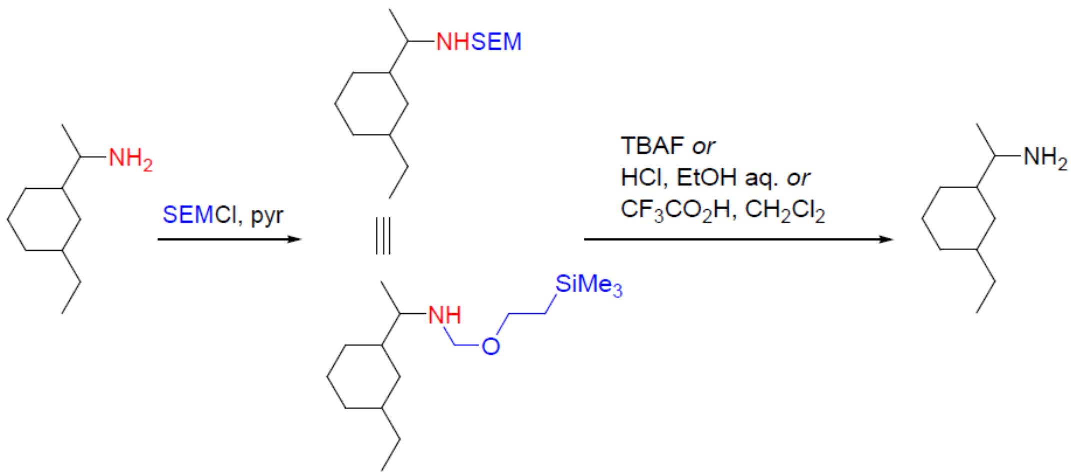
Formation:



Examples and cleavage conditions:



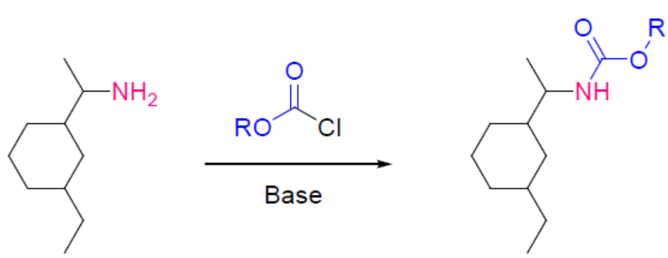




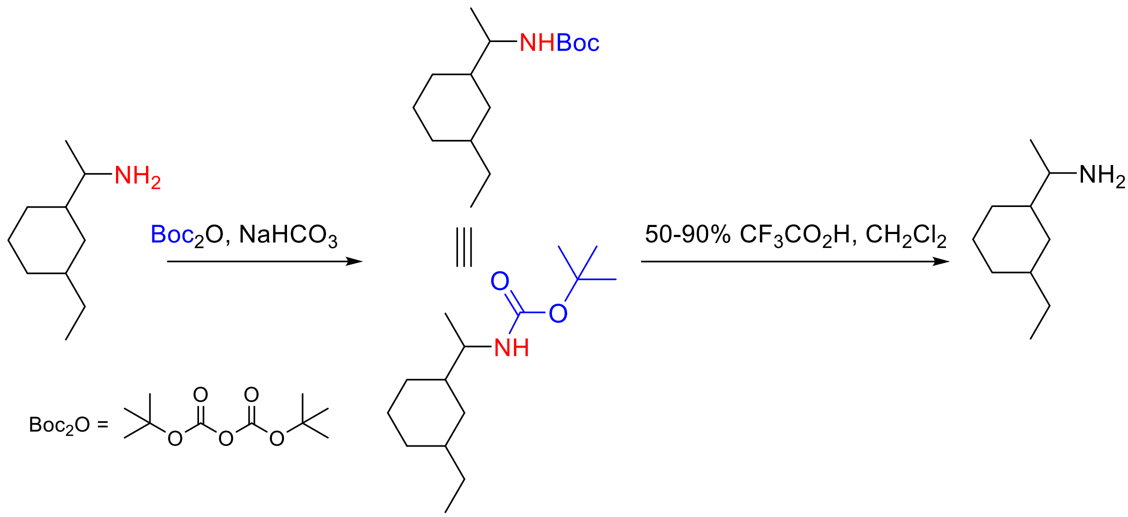
Carbamates

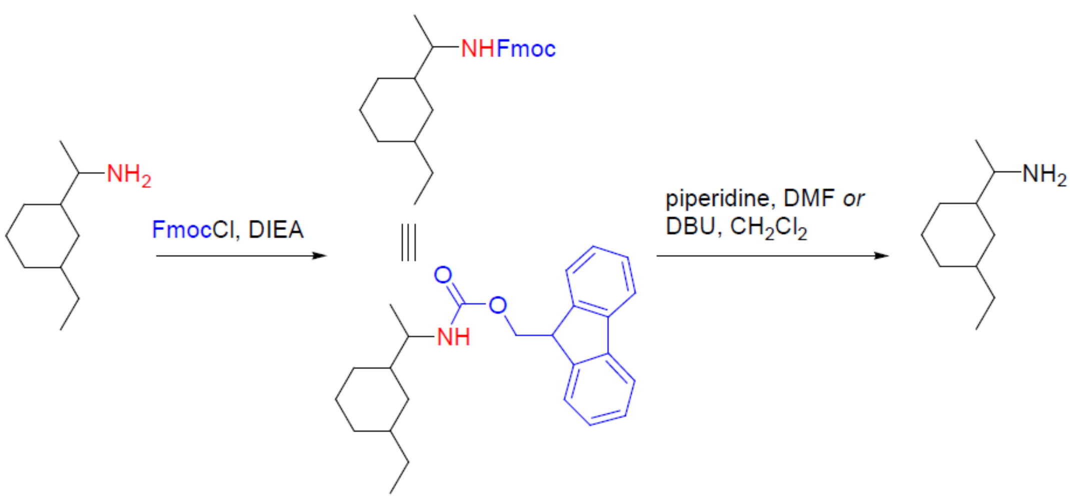
Protecting groups for primary and secondary amines

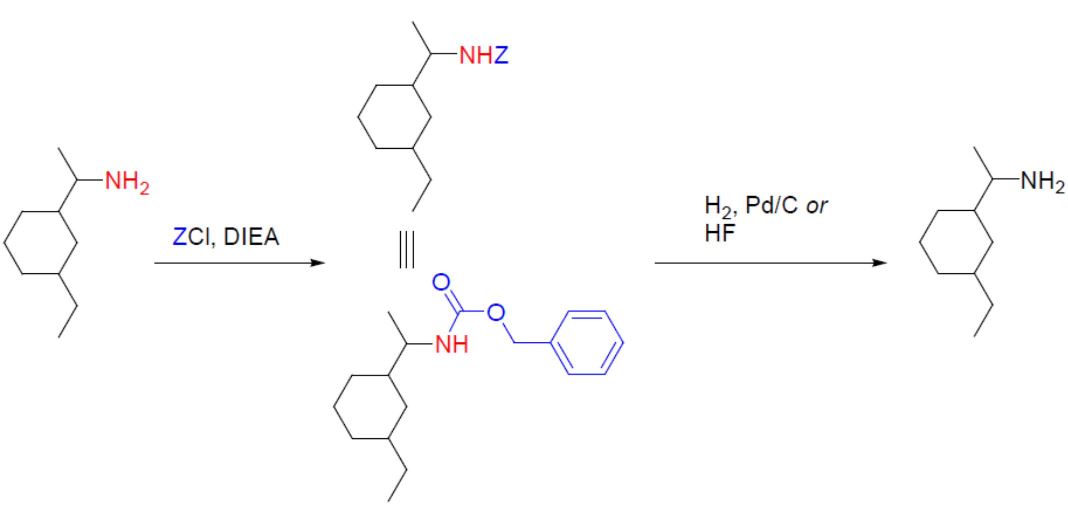
Formation:

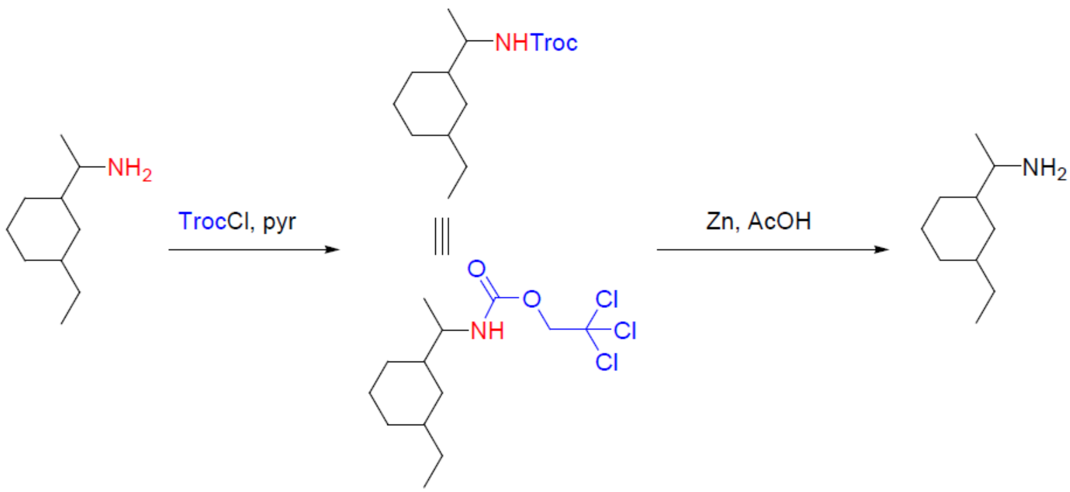


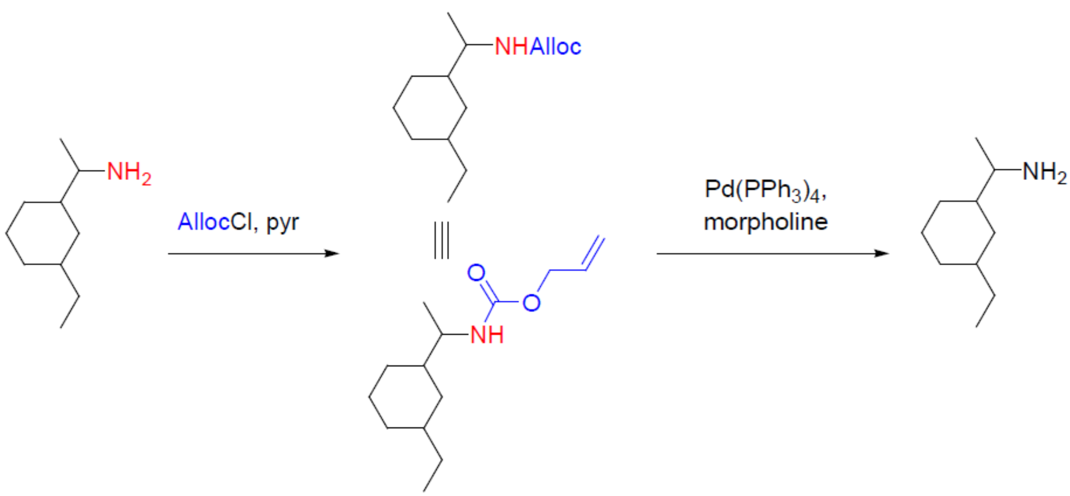
Examples and cleavage conditions:

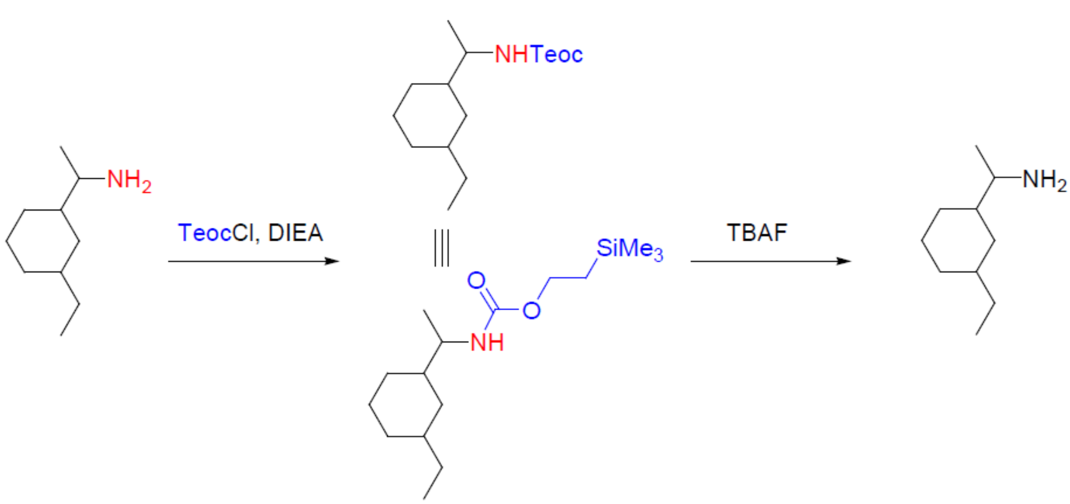







NH2
TrocCl, pyr
NHTroc
O
O
Cl
Cl
Cl
NH
Zn, AcOH
NH2


NH2
AllocCl, pyr
NHAlloc
O
O
NH
Pd(PPh3)4,
morpholine
NH2


NH2
TeocCl, DIEA
NHTeoc
SiMe3
O
O
NH
TBAF
NH2